

PRACTICAL-3

AIM: Implement Word Count / Frequency Program using MapReduce

PROBLEM DEFINITION:

A digital publishing company stores thousands of news articles and blog posts in a Hadoop Distributed File System (HDFS). The editorial team wants to identify the most frequently used words across their content to understand trending topics and optimize content recommendations. As a Big Data Engineer, you are required to develop a MapReduce application that processes the text data and generates word frequency statistics.

Task 1: Configure the Hadoop environment and upload the given text dataset to HDFS.

Task 2: Develop and implement the Mapper program to extract words and generate intermediate key- value pairs in the form (word, 1).

Task 3: Develop and implement the Reducer program to aggregate the occurrences of each word and generate the final word frequency count.

Task 4: Execute the MapReduce job on Hadoop and store the output in HDFS. Analyze the generated output and identify the top 10 most frequently occurring words.

Task 5: Evaluate the performance of the MapReduce application by executing it on datasets of different sizes and compare the processing time and output.

Supplementary Problem: Find the top 10 most frequently occurring words. Exclude stop words (e.g., “a,” “an,” “the,” “is”) from the analysis. Compare word frequency results for multiple input files.

SCREENSHOTS (SUPPLEMENTARY PROBLEMS INCLUDED):

```
(kali㉿kali)-[~/Desktop]
└─$ mkdir ~/wordcount

(kali㉿kali)-[~/Desktop]
└─$ cd ~/wordcount

(kali㉿kali)-[~/wordcount]
└─$ mkdir src

(kali㉿kali)-[~/wordcount]
└─$ mkdir input

(kali㉿kali)-[~/wordcount]
└─$ mkdir output

(kali㉿kali)-[~/wordcount]
└─$ nano input/news.txt

(kali㉿kali)-[~/wordcount]
└─$ cd ~/wordcount/src

(kali㉿kali)-[~/wordcount/src]
└─$ nano WordCountMapper.java
```

```
(kali㉿kali)-[~/Desktop]
└─$ mkdir ~/wordcount

(kali㉿kali)-[~/Desktop]
└─$ cd ~/wordcount

(kali㉿kali)-[~/wordcount]
└─$ mkdir src

(kali㉿kali)-[~/wordcount]
└─$ mkdir input

(kali㉿kali)-[~/wordcount]
└─$ mkdir output

(kali㉿kali)-[~/wordcount]
└─$ nano input/news.txt

(kali㉿kali)-[~/wordcount]
└─$ cd ~/wordcount/src

(kali㉿kali)-[~/wordcount/src]
└─$ nano WordCountMapper.java

(kali㉿kali)-[~/wordcount/src]
└─$ ls
WordCountMapper.java

(kali㉿kali)-[~/wordcount/src]
└─$ cd ~/WordCountProject/src
cd: no such file or directory: /home/kali/WordCountProject/src

(kali㉿kali)-[~/wordcount/src]
└─$ nano WordCountReducer.java

(kali㉿kali)-[~/wordcount/src]
└─$ nano WordCountDriver.java

(kali㉿kali)-[~/wordcount/src]
└─$ ls
WordCountDriver.java WordCountMapper.java WordCountReducer.java

(kali㉿kali)-[~/wordcount/src]
└─$ cd ~/WordCount
cd: no such file or directory: /home/kali/WordCount

(kali㉿kali)-[~/wordcount/src]
└─$ cd ..

(kali㉿kali)-[~/wordcount]
└─$ hadoop classpath
```

```

WordCountDriver.java WordCountMapper.java WordCountReducer.java
(kali@kali)-[~/wordcount/src]
$ cd ~/WordCount
cd: no such file or directory: /home/kali/WordCount
(kali@kali)-[~/wordcount/src]
$ cd ..
(kali@kali)-[~/wordcount]
$ hadoop classpath
/usr/local/hadoop/etc/hadoop:/usr/local/hadoop/share/hadoop/common/lib/*:/usr/local/hadoop/share/hadoop/common/*:/usr/local/hadoop/share/hadoop/hdfs:/usr/local/hadoop/share/hadoop/hdfs/lib/*:/usr/local/hadoop/share/hadoop/hdfs/*:/usr/local/hadoop/share/hadoop/mapreduce/*:/usr/local/hadoop/share/hadoop/yarn:/usr/local/hadoop/share/hadoop/yarn/lib/*:/usr/local/hadoop/share/hadoop/yarn/*
(kali@kali)-[~/wordcount]
$ javac -classpath "hadoop classpath" -d classes src/*.java
(kali@kali)-[~/wordcount]
$ find classes
classes
classes/WordCountReducer.class
classes/WordCountMapper.class
classes/WordCountDriver.class
(kali@kali)-[~/wordcount]
$ jar -cvf WordCount.jar -C classes/ .
added manifest
adding: WordCountDriver.class(in = 1539) (out= 852)(deflated 44%)
adding: WordCountMapper.class(in = 1938) (out= 885)(deflated 54%)
adding: WordCountReducer.class(in = 1735) (out= 745)(deflated 57%)
(kali@kali)-[~/wordcount]
$ ls
classes input output src WordCount.jar
(kali@kali)-[~/wordcount]
$ cd ~/WordCountProject/input
cd: no such file or directory: /home/kali/WordCountProject/input
(kali@kali)-[~/wordcount]
$ cd ~/input
cd: no such file or directory: /home/kali/input
(kali@kali)-[~/wordcount]
$ cd ~/WordCount/input
cd: no such file or directory: /home/kali/WordCount/input
(kali@kali)-[~/wordcount]
$ cd ~/WordCountProject/input
cd: no such file or directory: /home/kali/WordCountProject/input
(kali@kali)-[~/wordcount]
$ cd ~/input
cd: no such file or directory: /home/kali/input
(kali@kali)-[~/wordcount]
$ cd ~/WordCount/input
cd: no such file or directory: /home/kali/WordCount/input
(kali@kali)-[~/wordcount]
$ cd ~/input
cd: no such file or directory: /home/kali/input
(kali@kali)-[~/wordcount]
$ cd ..
(kali@kali)-[~/wordcount]
$ cd ~/home/kali/wordcount/input
(kali@kali)-[~/wordcount/input]
$ ls
classes input output src WordCount.jar
(kali@kali)-[~/wordcount/input]
$ nano news.txt
(kali@kali)-[~/wordcount/input]
$ hdfs dfs -mkdir -p /wordcount/input
2026-08-07 06:42:31,066 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
(kali@kali)-[~/wordcount/input]
$ cd ..
(kali@kali)-[~/wordcount]
$ cd ..
(kali@kali)-[~/wordcount]
$ hdfs dfs -mkdir -p /wordcount/input
2026-08-07 06:43:08,339 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

```

```

$ hdfs dfs -mkdir -p /wordcount/input
2026-08-07 06:43:08,339 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

(kali@kali)-[~]
$
(kali@kali)-[~]
$ hdfs dfs -ls /wordcount
2026-08-07 08:42:55,435 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
drwxr-xr-x - kali supergroup 0 2026-08-07 06:42 /wordcount/input

(kali@kali)-[~]
$ hdfs dfs -put ~/WordCountProject/input/news.txt /wordcount/input/
2026-08-07 08:43:45,456 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
put: '/home/kali/WordCountProject/input/news.txt': No such file or directory

(kali@kali)-[~]
$ ls ~/wordcount/input
news.txt

(kali@kali)-[~]
$ hdfs dfs -put ~/wordcount/input/news.txt /wordcount/input/
2026-08-07 08:44:59,758 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

(kali@kali)-[~]
$ hdfs dfs -ls /wordcount/input
2026-08-07 08:45:42,362 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r-- 1 kali supergroup 552 2026-08-07 08:45 /wordcount/input/news.txt

(kali@kali)-[~]
$ hadoop jar ~/WordCount/WordCount.jar WordCountDriver /wordcount/input /wordcount/output
2026-08-07 08:46:17,915 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-08-07 08:46:18,554 INFO client.DefaultHARMPFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-08-07 08:46:19,269 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner
2026-08-07 08:46:19,281 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1786093027586_0001
2026-08-07 08:46:20,018 INFO input.FileInputFormat: Total input files to process : 1
2026-08-07 08:46:20,151 INFO mapreduce.JobSubmitter: number of splits:1
2026-08-07 08:46:20,839 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1786093027586_0001
2026-08-07 08:46:20,840 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-08-07 08:46:21,144 INFO conf.Configuration: resource-types.xml not found
2026-08-07 08:46:21,145 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-08-07 08:46:21,854 INFO impl.YarnClientImpl: Submitted application application_1786093027586_0001
2026-08-07 08:46:21,912 INFO mapreduce.Job: The url to track the job: http://kali:8088/proxy/application_1786093027586_0001/
2026-08-07 08:46:21,913 INFO mapreduce.Job: Running job: job_1786093027586_0001
2026-08-07 08:46:27,134 INFO mapreduce.Job: Job job_1786093027586_0001 running in uber mode : false
2026-08-07 08:46:27,157 INFO mapreduce.Job: Job job_1786093027586_0001 failed with state FAILED due to: Application application_1786093027586_0001 failed 2 times due to AM Container
exitCode: 1
Failing this attempt.Diagnostics: [2026-08-07 08:46:27,050]Exception from container-launch.
Container id: container_1786093027586_0001_02_000001
Exit code: 1

[2026-08-07 08:46:27,077]Container exited with a non-zero exit code 1. Error file: prelaunch.err.
Last 4096 bytes of prelaunch.err :
Last 4096 bytes of stderr :
Error: Could not find or load main class org.apache.hadoop.mapreduce.v2.app.MRAppMaster
Caused by: java.lang.ClassNotFoundException: org.apache.hadoop.mapreduce.v2.app.MRAppMaster

Please check whether your <HADOOP_HOME>/etc/hadoop/mapred-site.xml contains the below configuration:
<property>
<name>yarn.app.mapreduce.am.env</name>
<value>HADOOP_MAPRED_HOME=${full path of your hadoop distribution directory}</value>
</property>
<property>
<name>mapreduce.map.env</name>
<value>HADOOP_MAPRED_HOME=${full path of your hadoop distribution directory}</value>
</property>
<property>
<name>mapreduce.reduce.env</name>
<value>HADOOP_MAPRED_HOME=${full path of your hadoop distribution directory}</value>
</property>

[2026-08-07 08:46:27,077]Container exited with a non-zero exit code 1. Error file: prelaunch.err.
Last 4096 bytes of prelaunch.err :
Last 4096 bytes of stderr :
Error: Could not find or load main class org.apache.hadoop.mapreduce.v2.app.MRAppMaster
Caused by: java.lang.ClassNotFoundException: org.apache.hadoop.mapreduce.v2.app.MRAppMaster

Please check whether your <HADOOP_HOME>/etc/hadoop/mapred-site.xml contains the below configuration:
<property>
(kali@kali)-[~]
$ ls /usr/local/hadoop/etc/hadoop/
capacity-scheduler.xml hadoop-env.cmd hadoop-user-functions.sh.example https-log4j.properties kms-log4j.properties mapred-env.sh ssl-client.xml.example yarn-env.cmd
configuration.xml hadoop-env.sh hadfs-site.xml https-site.xml kms-site.xml kms-site.xml mapred-queues.xml.template ssl-server.xml.example yarn-env.sh
container-executor.cfg hadoop-metrics2.properties hadfs-site.xml kms-acls.xml kms-site.xml log4j.properties mapred-site.xml user_ec_policies.xml.template yarnservice-log4j.properties
core-site.xml hadoop-policy.xml https-env.sh kms-env.sh mapred-env.cmd shellprofile.d yarn-site.xml

(kali@kali)-[~]
$ nano /usr/local/hadoop/etc/hadoop/mapred-site.xml

(kali@kali)-[~]
$ echo $HADOOP_MAPRED_HOME
/usr/local/hadoop

(kali@kali)-[~]
$ stop-yarn.sh
stop-dfs.sh
Stopping nodemanagers
localhost: WARNING: nodemanager did not stop gracefully after 5 seconds: Trying to kill with kill -9
Stopping resourcemanager
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [kali]
2026-08-07 08:51:42,891 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

(kali@kali)-[~]
$ start-dfs.sh
start-yarn.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [kali]
2026-08-07 08:52:12,247 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Starting nodemanagers

(kali@kali)-[~]
$ jps
\
117081 Jps
116739 SecondaryNameNode
116388 NameNode
117128 NodeManager
115086 DataNode
117084 ResourceManager

(kali@kali)-[~]
$ \
> hdfs dfs -ls /wordcount/output
2026-08-07 08:53:14,380 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
ls: '/wordcount/output': No such file or directory

```

```

(kali@kali)-[~]
$ hdfs dfs -rm -r /wordcount/output
2026-09-07 09:53:54,240 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
rm: '/wordcount/output': No such file or directory

(kali@kali)-[~]
$ echo $HADOOP_HOME

echo $HADOOP_MAPRED_HOME

cat /usr/local/hadoop/etc/hadoop/mapred-site.xml

jps
/usr/local/hadoop
/usr/local/hadoop
<?xml version="1.0"?>

<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>

  <property>
    <name>yarn.app.mapreduce.am.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>

  <property>
    <name>mapreduce.map.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>

  <property>
    <name>mapreduce.reduce.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>
</configuration>
116739 SecondaryNameNode
118597 Jps
116388 NameNode
117128 NodeManager
116506 DataNode
117004 ResourceManager

(kali@kali)-[~]
$ nano /usr/local/hadoop/etc/hadoop/mapred-site.xml

(kali@kali)-[~]
$ ls /usr/local/hadoop/share/hadoop/mapreduce
hadoop-mapreduce-client-app-3.4.1.jar  hadoop-mapreduce-client-hs-3.4.1.jar  hadoop-mapreduce-client-jobclient-3.4.1-tests.jar  hadoop-mapreduce-client-uploader-3.4.1.jar  sources
hadoop-mapreduce-client-common-3.4.1.jar  hadoop-mapreduce-client-hs-plugins-3.4.1.jar  hadoop-mapreduce-client-native-task-3.4.1.jar  hadoop-mapreduce-examples-3.4.1.jar
hadoop-mapreduce-client-core-3.4.1.jar  hadoop-mapreduce-client-jobclient-3.4.1.jar  hadoop-mapreduce-client-shuffle-3.4.1.jar  jdiff

(kali@kali)-[~]
$ hadoop version
Hadoop 3.4.1
Source code repository https://github.com/apache/hadoop.git -r 4d7825309348956336b8f06a08322b78422849b1
Compiled by mthakur on 2024-10-09T14:57Z
Compiled on platform linux-x86_64
Compiled with protoc 3.23.4
From source with checksum 7292fe9dba5e2e44e3a9f763f3e3e680
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3.4.1.jar

(kali@kali)-[~]
$ cat /usr/local/hadoop/etc/hadoop/yarn-site.xml
<?xml version="1.0"?>
<!--
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you may not use this file except in compliance with the License.
You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<configuration>
<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>
<!-- Site specific YARN configuration properties -->
</configuration>

```

```
(kali@kali)-[~]
$ cat /usr/local/hadoop/etc/hadoop/yarn-site.xml
<?xml version="1.0"?>
<!--
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you may not use this file except in compliance with the License.
You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

Unless required by applicable law or agreed to in writing, software
distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License. See accompanying LICENSE file.
-->
<configuration>
<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>
<!-- Site specific YARN configuration properties -->
</configuration>

(kali@kali)-[~]
$ cat /usr/local/hadoop/etc/hadoop/hadoop-env.sh | grep JAVA_HOME
# JAVA_HOME=/usr/java/testing hdfs dfs -ls
# Technically, the only required environment variable is JAVA_HOME.
# export JAVA_HOME=
export JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64

(kali@kali)-[~]
$ cat /usr/local/hadoop/etc/hadoop/mapred-env.sh | grep JAVA_HOME

(kali@kali)-[~]
$ nano /usr/local/hadoop/etc/hadoop/yarn-site.xml
```

```
(kali@kali)-[~]
$ cat /usr/local/hadoop/etc/hadoop/mapred-env.sh | grep JAVA_HOME

(kali@kali)-[~]
$ nano /usr/local/hadoop/etc/hadoop/yarn-site.xml

(kali@kali)-[~]
$ cat /usr/local/hadoop/etc/hadoop/mapred-site.xml
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
  <property>
    <name>yarn.app.mapreduce.am.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>
  <property>
    <name>mapreduce.map.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>
  <property>
    <name>mapreduce.reduce.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>
</configuration>

(kali@kali)-[~]
$ stop-yarn.sh
stop-dfs.sh
Stopping nodemanagers
Stopping resourcemanager
Stopping namenodes on [localhost]
Stopping datanodes
Stopping secondary namenodes [kali]
2026-08-07 09:02:40,578 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
```

```
(kali@kali)-[~]
$ start-dfs.sh
start-yarn.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [kali]
2026-08-07 09:03:02,514 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting resourcemanager
Starting nodemanagers

(kali@kali)-[~]
$ hdfs dfs -rm -r /wordcount/output
2026-08-07 09:03:25,486 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
rm: '/wordcount/output': No such file or directory

(kali@kali)-[~]
$ hadoop jar ~/wordcount/WordCount.jar WordCountDriver /wordcount/input /wordcount/output
2026-08-07 09:03:35,782 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-08-07 09:03:36,303 INFO client.DefaultHadoopFailoverProxyProvider: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-08-07 09:03:36,636 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-08-07 09:03:36,685 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1786107789794_0001
2026-08-07 09:03:37,203 INFO input.FileInputFormat: Total input files to process : 1
2026-08-07 09:03:38,171 INFO mapreduce.JobSubmitter: number of splits:1
2026-08-07 09:03:38,370 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1786107789794_0001
2026-08-07 09:03:38,370 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-08-07 09:03:38,553 INFO conf.Configuration: resource-types.xml not found
2026-08-07 09:03:38,553 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-08-07 09:03:39,122 INFO impl.YarnClientImpl: Submitted application application_1786107789794_0001
2026-08-07 09:03:39,199 INFO mapreduce.Job: The url to track the job: http://kali:8088/proxy/application_1786107789794_0001/
2026-08-07 09:03:39,200 INFO mapreduce.Job: Running job: job_1786107789794_0001
2026-08-07 09:03:40,893 INFO mapreduce.Job: Job job_1786107789794_0001 running in uber mode : false
2026-08-07 09:03:40,895 INFO mapreduce.Job: map 0% reduce 0%
2026-08-07 09:03:53,971 INFO mapreduce.Job: map 100% reduce 0%
2026-08-07 09:03:59,134 INFO mapreduce.Job: map 100% reduce 100%
2026-08-07 09:04:00,266 INFO mapreduce.Job: Job job_1786107789794_0001 completed successfully
2026-08-07 09:04:00,418 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=988
  FILE: Number of bytes written=619575
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=663
  HDFS: Number of bytes written=440
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=2726
  Total time spent by all reduces in occupied slots (ms)=2945
  Total time spent by all map tasks (ms)=2726
  Total time spent by all reduce tasks (ms)=2945
  Total vcore-milliseconds taken by all map tasks=2726
  Total vcore-milliseconds taken by all reduce tasks=2945
  Total megabyte-milliseconds taken by all map tasks=2791424
  Total megabyte-milliseconds taken by all reduce tasks=3015680
Map-Reduce Framework
  Map input records=23
  Map output records=75
  Map output bytes=832
  Map output materialized bytes=988
  Input split bytes=111
```

```
(kali@kali)-[~]
$ ls /usr/local/hadoop/etc/hadoop/mapred-env.sh
/usr/local/hadoop/etc/hadoop/mapred-env.sh

(kali@kali)-[~]
$ cat /usr/local/hadoop/etc/hadoop/mapred-env.sh
# Licensed to the Apache Software Foundation (ASF) under one or more
# contributor license agreements. See the NOTICE file distributed with
# this work for additional information regarding copyright ownership.
# The ASF licenses this file to You under the Apache License, Version 2.0
# (the "License"); you may not use this file except in compliance with
# the License. You may obtain a copy of the License at
#
# http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
# See the License for the specific language governing permissions and
# limitations under the License.
##
## THIS FILE ACTS AS AN OVERRIDE FOR hadoop-env.sh FOR ALL
## WORK DONE BY THE mapred AND RELATED COMMANDS.
##
## Precedence rules:
##
## mapred-env.sh > hadoop-env.sh > hard-coded defaults
##
## MAPRED_xyz > HADOOP_xyz > hard-coded defaults
##
###
# Job History Server specific parameters
###
# Specify the max heapsize for the JobHistoryServer. If no units are
# given, it will be assumed to be in MB.
# This value will be overridden by an Xmx setting specified in HADOOP_OPTS,
# and/or MAPRED_HISTORYSERVER_OPTS.
# Default is the same as HADOOP_HEAPSIZE_MAX.
#export HADOOP_JOB_HISTORYSERVER_HEAPSIZE=
# Specify the JVM options to be used when starting the HistoryServer.
# These options will be appended to the options specified as HADOOP_OPTS
# and therefore may override any similar flags set in HADOOP_OPTS
#export MAPRED_HISTORYSERVER_OPTS=
# Specify the log4j settings for the JobHistoryServer
# Java property: hadoop.root.logger
```

```

(kali@kali)~$ nano /usr/local/hadoop/etc/hadoop/mapred-env.sh
(kali@kali)~$ cat /usr/local/hadoop/etc/hadoop/mapred-site.xml
<?xml version="1.0"?>

<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>

<configuration>

  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>

  <property>
    <name>yarn.app.mapreduce.am.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>

  <property>
    <name>mapreduce.map.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>

  <property>
    <name>mapreduce.reduce.env</name>
    <value>HADOOP_MAPRED_HOME=/usr/local/hadoop</value>
  </property>

</configuration>

(kali@kali)~$ hdfs dfs -ls /wordcount/output
2026-08-07 09:06:42,066 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 kali supergroup          0 2026-08-07 09:03 /wordcount/output/_SUCCESS
-rw-r--r-- 1 kali supergroup    440 2026-08-07 09:03 /wordcount/output/part-r-00000

(kali@kali)~$ hdfs dfs -cat /wordcount/output/part-r-00000
2026-08-07 09:06:54,532 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
a 1
across 1
an 1
analytics 1
and 4
artificial 1
better 1
big 5
data 10
datasets 1
decisions 1
distributed 1
fault 1
for 2
framework 1
hadoop 5
hdfs 1
helps 1
intelligence 1
involves 1
is 2
large 1
learning 5
machine 2
machines 1
make 1
mapreduce 3
model 1
multiple 1
of 1
open 1
organizations 1
processing 2
programming 1
provides 1
scalability 1
science 1
source 1
statistics 1
storage 1
stores 1
tolerance 1
use 1
used 1
visualization 1

```

```

(kali@kali)-[~]
$ mkdir ~/wordcount/result
hdfs dfs -get /wordcount/output/part-r-00000 ~/wordcount/result/
2026-08-07 09:07:19,851 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

(kali@kali)-[~]
$ cat ~/wordcount/result/part-r-00000
a 1
across 1
an 1
analytics 1
and 4
artificial 1
better 1
big 5
data 10
datasets 1
decisions 1
distributed 1
fault 1
for 2
framework 1
hadoop 5
hdfs 1
helps 1
intelligence 1
involves 1
is 2
large 1
learning 5
machine 2
machines 1
make 1
mapreduce 3
model 1
multiple 1
of 1
open 1
organizations 1
processing 2
programming 1
provides 1
scalability 1
science 1
source 1
statistics 1
storage 1
stores 1
tolerance 1
use 1
used 1

(kali@kali)-[~]
$ hdfs dfs -rm -r /wordcount/output
2026-08-07 09:07:44,333 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /wordcount/output

(kali@kali)-[~]
$ time hadoop jar ~/wordcount/WordCount.jar WordCountDriver /wordcount/input /wordcount/output
2026-08-07 09:07:55,612 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-08-07 09:07:56,132 INFO client.DefaultHadoopFileOverProxyProvider: Connecting to ResourceManager at localhost/127.0.0.1:8032
2026-08-07 09:07:56,428 INFO mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-08-07 09:07:56,465 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/kali/.staging/job_1786107789794_0002
2026-08-07 09:07:56,861 INFO input.FileInputFormat: Total input files to process : 1
2026-08-07 09:07:56,965 INFO mapreduce.JobSubmitter: number of splits:1
2026-08-07 09:07:57,635 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1786107789794_0002
2026-08-07 09:07:57,635 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-08-07 09:07:57,849 INFO conf.Configuration: resource-types.xml not found
2026-08-07 09:07:57,850 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-08-07 09:07:57,929 INFO impl.YarnClientImpl: Submitted application application_1786107789794_0002
2026-08-07 09:07:57,964 INFO mapreduce.Job: The url to track the job: http://kali:8088/proxy/application_1786107789794_0002/
2026-08-07 09:07:57,966 INFO mapreduce.Job: Running job: job_1786107789794_0002
2026-08-07 09:08:04,167 INFO mapreduce.Job: Job job_1786107789794_0002 running in uber mode : false
2026-08-07 09:08:04,170 INFO mapreduce.Job: map 0% reduce 0%
2026-08-07 09:08:09,287 INFO mapreduce.Job: map 100% reduce 0%
2026-08-07 09:08:15,407 INFO mapreduce.Job: map 100% reduce 100%
2026-08-07 09:08:17,459 INFO mapreduce.Job: Job job_1786107789794_0002 completed successfully
2026-08-07 09:08:17,545 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=988
  FILE: Number of bytes written=619575
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=663
  HDFS: Number of bytes written=440
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=2834
  Total time spent by all reduces in occupied slots (ms)=3305
  Total time spent by all map tasks (ms)=2834
  Total time spent by all reduce tasks (ms)=3305
  Total vcore-milliseconds taken by all map tasks=2834
  Total vcore-milliseconds taken by all reduce tasks=3305
  Total megabyte-milliseconds taken by all map tasks=2902016
  Total megabyte-milliseconds taken by all reduce tasks=3384320
Map-Reduce Framework
  Map input records=23
  Map output records=75
  Map output bytes=832
  Map output materialized bytes=988

```

```

Total megabyte-milliseconds taken by all reduce tasks=3384320
Map-Reduce Framework
  Map input records=23
  Map output records=75
  Map output bytes=832
  Map output materialized bytes=988
  Input split bytes=111
  Combine input records=0
  Combine output records=0
  Reduce input groups=45
  Reduce shuffle bytes=988
  Reduce input records=75
  Reduce output records=45
  Spilled Records=150
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=43
  CPU time spent (ms)=1290
  Physical memory (bytes) snapshot=477925376
  Virtual memory (bytes) snapshot=5490798592
  Total committed heap usage (bytes)=252706816
  Peak Map Physical memory (bytes)=290226176
  Peak Map Virtual memory (bytes)=2742767616
  Peak Reduce Physical memory (bytes)=187699200
  Peak Reduce Virtual memory (bytes)=2748030976
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=552
File Output Format Counters
  Bytes Written=440

real    22.75s
user    2.49s
sys     1.92s
cpu     19%

(kali@kali)-[~]
$

```

KEY QUESTIONS / ANALYSIS:

1) What is the role of Mapper and Reducer in the Word Count application?

Mapper:

The Mapper reads the input text, splits it into individual words, and generates intermediate keyvalue pairs in the form (word, 1).

Reducer:

The Reducer receives all values associated with the same word and adds them to calculate the total frequency.

2) How are intermediate key-value pairs generated and processed?

The Mapper converts every word into:

(word, 1)

Hadoop then performs shuffle and sort, which groups identical keys together.

For example:

(hadoop, 1)

(hadoop, 1)

(hadoop, 1)

becomes:

hadoop $\rightarrow$ [1, 1, 1]

The Reducer then calculates:

hadoop $\rightarrow$ 3

3) How does HDFS store and distribute the input dataset?

HDFS divides large files into blocks and stores these blocks across different DataNodes in the Hadoop cluster.

It also maintains multiple copies of blocks using replication, which provides fault tolerance.

The NameNode manages metadata and information about where the blocks are stored, while DataNodes store the actual data blocks.

4) What advantages does MapReduce offer for processing large datasets?

MapReduce provides:

- Parallel processing
- Scalability
- Fault tolerance
- Distributed processing
- Efficient handling of large datasets
- Data locality, which allows processing to occur close to where data is stored.

5) How does execution time vary with dataset size?

Generally, execution time increases as the dataset size increases because more data must be read, processed, shuffled, and reduced.

However, MapReduce can distribute the workload across multiple nodes, allowing large datasets to be processed efficiently.

As the dataset size increases, the benefits of parallel processing become more significant.

CONCLUSION:

The Word Count/Frequency program was successfully implemented using Hadoop MapReduce. The Mapper generated intermediate key-value pairs in the form (word, 1), while the Reducer aggregated these values to calculate the frequency of each word. The input data was stored and processed using HDFS, demonstrating distributed storage and parallel processing. The top 10 frequently occurring words were identified, and the performance was evaluated using datasets of different sizes. Overall, MapReduce provides an efficient and scalable approach for processing large text datasets.